

# Countdown 31 — Master Backend Engineering Specification

Comprehensive, unified architectural master document and implementation contract covering all 7 engineering volumes: Base Game Engine, Options & Profile, Web3 Transactions, Store Economy, Tournaments Platform, Sponsor Campaign Studio, and Admin Security & DevOps.

**Target Stack:** NestJS 11 · Prisma 6 · PostgreSQL 16 · Redis 7 · Socket.IO · Solana Web3 · Next.js 16  
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**EXECUTIVE MASTER DIRECTIVE:**  
*This Master Specification provides complete technical alignment between frontend and backend engineering. It guarantees that every screen, button, hook, WebSocket event, financial transaction, sponsor template, and administrative workflow in the repository operates seamlessly in high-concurrency production.*

## Master Specification Index & Volume Structure

| Volume   | Subject & Focus Area                 | Key Modules, Models & Protocols Covered                                                |
|----------|--------------------------------------|----------------------------------------------------------------------------------------|
| Volume 1 | Base Game Engine & Multiplayer       | Counting 1-31, 7s turn timer, Skills mode, Socket.IO gateway, AI bots, Barnaby mascot. |
| Volume 2 | Options, Settings, Profile & Avatars | User stats, match history, 6-layer avatar studio, TOTP 2FA, session revocation.        |
| Volume 3 | Web3 Wallet & USDT Ledger            | Solana SPL USDT deposits, DepositBinding, double-entry ledger, prize pool escrow.      |
| Volume 4 | Store, Cosmetics & Economy           | Shop catalog, atomic purchase flow, cosmeticsOwned JSON, equip validation.             |
| Volume 5 | Tournaments & Influencer Battles     | Solo & team battles, gate codes (sponsor/join/captain/member), GMT time voting.        |
| Volume 6 | Campaign Studio & Telemetry          | Draft versioning, BRAND/SAFETY reviews, [Sponsor] 31 branding, templates, telemetry.   |
| Volume 7 | Admin Console & DevOps               | User moderation, sponsor provisioning, rate-limiting, Docker Compose, CI/CD.           |

## Complete REST API Catalogue & Routing Index

| Domain / Tag     | Endpoint & Method                                                                    | Auth Guard & Capabilities                                                           |
|------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Auth & Security  | POST /auth/login<br>POST /auth/register<br>POST /auth/mfa/setup                      | JWT access/refresh tokens; TOTP 2FA setup with encrypted base32 secrets.            |
| Tournaments      | GET /promo-tournaments<br>POST /promo-tournaments/:id/join<br>POST /...:id/vote-time | Public discovery, referral join validation, GMT start-time slot voting.             |
| Campaign Studio  | GET /admin/.../campaign<br>PATCH /admin/.../draft<br>POST /admin/.../publish         | Draft versioning, BRAND/SAFETY review decisions, 1-click historical rollback.       |
| Templates Engine | GET /admin/campaign-templates<br>POST /admin/.../apply-template                      | Global presets (Nike 31, Adidas 31, etc.) and atomic 1-click tournament deployment. |
| Telemetry        | POST /.../campaign/events<br>GET /admin/.../report/csv                               | Privacy-safe impression batch ingestion and proof-of-performance CSV streamer.      |
| Wallet & Shop    | GET /wallet/balance<br>POST /shop/purchase<br>PATCH /me/cosmetics                    | Solana USDT ledger balances, atomic shop purchases, 6-layer cosmetic equips.        |

## Developer Execution & Quality Assurance Checklist

- \*\*Prisma Migrations:\*\*** Apply `schema.prisma` migrations covering all 10 core entity tables.
- \*\*Seeding:\*\*** Run `prisma/seed.ts` to initialize treasury funds (1,000,000 USDT), admin, demo players, and 7 presets.
- \*\*Gateways & Sockets:\*\*** Wire Redis adapter to `CountdownGateway` on `/countdown` namespace.
- \*\*Telemetry Worker:\*\*** Spin up BullMQ worker to aggregate 10-second client telemetry into PostgreSQL.
- \*\*End-to-End Testing:\*\*** Run test suite verifying 100% contract adherence with Next.js frontend client.